

Time on the Cross

The Economics
of American Negro Slavery

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Prologue.

Slavery and the Cliometric Revolution

The years of black enslavement and the Civil War in which they terminated were our nation's time on the cross. Certainly if any aspect of the American past evokes a sense of shame, it is the system of slavery. And no war in our history, including World Wars I and II, took more American lives, was more devastating to our economy, or more threatening to our national survival, than the Civil War. The desire of scholars to lay bare the economic, political, and social forces which produced the tensions of the antebellum era and exploded into the worst holocaust of our history is not difficult to appreciate.

A vast literature has accumulated on the operation of the slave economy. Hundreds of historians have examined virtually every aspect of the institution. While scholars have disagreed on many points, a broad consensus has gradually emerged. By the midpoint of the twentieth century the consensus had been repeated so often that it had earned the status of a traditional interpretation. This interpretation is taught in most high school and college classes across the nation. It is the view that has been accepted by most of the readers of this book.

What is not generally known is that the traditional interpretation of slavery has been under intensive critical review for almost a decade and a half by historians and economists who are trained in the application of quantitative methods to historical problems. This review involves the processing of large quantities of numerical data. Although these data have been available for some time, the techniques required to analyze and interpret them systematically were not perfected until after the close of World War II. Then a series of rapid advances in economics, statistics, and applied mathematics, together with the availability of high-speed computers, put information long locked in obscure archives at the disposal of a new generation of scholars.

The review based on these new techniques and hitherto neglected sources has contradicted many of the most important propositions in the traditional portrayal of the slave system. As significant as the correction of past errors is the new information brought to light on the conditions of black bondage. Though the investigations are still in progress, enough have been completed so that the main features of the actual operation of the slave economy are now clear. The reconstruction which has emerged is so much at variance with common beliefs, and its implications are so central to the understanding of contemporary issues, that we believe the new findings should no longer be restricted to the pages of esoteric scholarly journals.

The following are some of the principal corrections of the traditional characterization of the slave economy:

1. Slavery was not a system irrationally kept in existence by plantation owners who failed to perceive or were indifferent to their best economic interests. The purchase of a slave was generally a highly profitable investment which yielded rates of return that compared favorably with the most outstanding investment opportunities in manufacturing.
2. The slave system was not economically moribund on the

eve of the Civil War. There is no evidence that economic forces alone would have soon brought slavery to an end without the necessity of a war or some other form of political intervention. Quite the contrary; as the Civil War approached, slavery as an economic system was never stronger and the trend was toward even further entrenchment.

3. Slaveowners were not becoming pessimistic about the future of their system during the decade that preceded the Civil War. The rise of the secessionist movement coincided with a wave of optimism. On the eve of the Civil War, slaveholders anticipated an era of unprecedented prosperity.

4. Slave agriculture was not inefficient compared with free agriculture. Economies of large-scale operation, effective management, and intensive utilization of labor and capital made southern slave agriculture 35 percent more efficient than the northern system of family farming.

5. The typical slave field hand was not lazy, inept, and unproductive. On average he was harder-working and more efficient than his white counterpart.

6. The course of slavery in the cities does not prove that slavery was incompatible with an industrial system or that slaves were unable to cope with an industrial regimen. Slaves employed in industry compared favorably with free workers in diligence and efficiency. Far from declining, the demand for slaves was actually increasing more rapidly in urban areas than in the countryside.

7. The belief that slave-breeding, sexual exploitation, and promiscuity destroyed the black family is a myth. The family was the basic unit of social organization under slavery. It was to the economic interest of planters to encourage the stability of slave families and most of them did so. Most slave sales were either of whole families or of individuals who were at an age when it would have been normal for them to have left the family.

8. The material (not psychological) conditions of the lives of slaves compared favorably with those of free industrial workers. This is not to say that they were good by modern standards. It merely emphasizes the hard lot of all workers, free or slave, during the first half of the nineteenth century.

9. Slaves were exploited in the sense that part of the income which they produced was expropriated by their owners. However, the rate of expropriation was much lower than has generally been presumed. Over the course of his lifetime, the typical slave

field hand received about 90 percent of the income he produced. 10. Far from stagnating, the economy of the antebellum South grew quite rapidly. Between 1840 and 1860, per capita income increased more rapidly in the South than in the rest of the nation. By 1860 the South attained a level of per capita income which was high by the standards of the time. Indeed, a country as advanced as Italy did not achieve the same level of per capita income until the eve of World War II.

Even this partial summary of corrections raises the question of how those who fashioned the traditional interpretation of the slave system could have been so wrong. The historians whose views are now under criticism were conscientious and diligent scholars; they were highly intelligent and perceptive; and they strove to portray the history of the antebellum South as it actually was. The explanation of how such men were misled does not turn on issues of personal bias or other idiosyncratic behavior — even if, in some instances, bias has been a contributing factor. It hinges to a large extent on certain broad methodological questions, and particularly on the role of mathematics and statistics in historical analysis. For some of the most telling revisions of the new work turn on technical mathematical points, points which despite their obscurity are vital to the correct description and interpretation of the slave economy.

Antebellum slavery is not the only aspect of American economic history which has undergone radical revision as a result of the application of mathematical and statistical methods. The reexamination of slavery is part of a more ambitious effort to reconstruct the entire history of American economic development on a sound quantitative basis. Those engaged in this enterprise are called “new economic historians,” “econometric historians,” and “cliometricians.” The first definitely formulated expression of the new approach was set forth in a pair of essays written in 1957 by Alfred H. Conrad and John R. Meyer, who were then young assistant professors at Harvard. Since the appearance of these

papers, the number of scholars attempting to apply mathematical and statistical methods to history increased from less than a score to several hundred. As their numbers have proliferated so have their iconoclastic findings.

The cliometricians have downgraded the role of technological change in American economic advance; they have controverted the claim that railroads were necessary to the settlement and exploitation of the West; they have contended that the boom and bust of the 1830s and early 1840s were the consequences of developments in Mexico and Britain rather than the policies of Andrew Jackson; and they have rejected the contention that the Civil War greatly accelerated the industrialization of the nation.

Today the findings and methods of the new economic history are routinely taught at the graduate level at Harvard, Yale, Chicago, the University of California, and other leading universities which together produce most of the Ph.D.s in the field. This does not mean that the controversy over the cliometric approach has come to an end. There are still a significant number of eminent economic historians who are deeply skeptical of the usefulness of mathematical methods.

But even the most skeptical critics of the “mathematical craze” have had to concede that the cliometricians have performed a service in searching out and systematically sifting through huge quantities of data of direct relevance to the understanding of U.S. economic development. And nowhere has this passion for numbers been more vigorously pursued than in the study of the economics of slavery.

For more than a decade several scores of scholars have been exploring every conceivable source of information bearing on the operation of the slave system. The search led them into the deepest recesses of the National Archives and of various state archives where the original, handwritten schedules of information collected by the census takers of 1850 and 1860 were stored. These documents contain quite detailed information about every plantation then extant as

well as about the slaves who were born, lived, and died in those years. The cliometricians also scoured the papers of the historical societies of various southern states and some states in the North, where are deposited the family papers and business records of the largest planters. The wills and other legal documents of the estates of planters have been particularly valuable. In addition to yielding prices on tens of thousands of slaves, these records have been one of the principal bases for determination of the structural characteristics of black families. Pursuit of this body of evidence eventually led even to such places as the Wasatch Mountains of Utah, where the Genealogical Society of the Church of Jesus Christ of Latter-Day Saints has stored microfilms of probate records gathered from hundreds of county court-houses.

As a result of the search, the cliometricians have amassed a more complete body of information on the operation of the slave system than has been available to anyone interested in the subject either during the antebellum era or since then. It is this enormous body of evidence which is the source of many of their new discoveries.

Some of the discoveries were at one time as unbelievable to the cliometricians as they will be to the readers of this volume. Indeed, many of the findings presented in the chapters that follow were initially discounted, even rejected out of hand. But when persistent efforts to contradict the unexpected discoveries failed, these scholars were forced into a wide-ranging and radical reinterpretation of American slavery.

This will be a disturbing book to read. It requires forbearance on the part of the reader and a recognition that what is set forth represents the honest efforts of scholars whose central aim has been the discovery of what really happened. We believe that this forbearance will prove worthwhile. For the findings we discuss not only expose many myths that have served to corrode and poison relations

between the races, but also help to put into a new perspective some of the most urgent issues of our day.

In considering the evidence presented in this book, readers should keep in mind certain caveats regarding the advantages and limitations of cliometric methods. The advantage of the application of quantitative methods to history is not that it provides unambiguous answers to all questions. Not all questions have unambiguous answers. And many of those questions which in principle have unambiguous answers cannot be resolved because of the absence of crucial bodies of data, because the retrieval of some bodies of data are too expensive to be practical, or because the analysis of a given body of data poses problems that cannot be treated by the mathematical and statistical methods that have thus far been developed.

It is, for example, much easier to obtain data bearing on the frequency with which slave families were broken up by the interregional slave trade than on the inner quality of the family life of slaves. Similarly, the evidence on the number of slaves per slave house is more direct and complete than the evidence available on the cleanliness and healthfulness of that housing. And, while the cliometricians have been able to construct reasonably reliable indexes of the material level at which blacks lived under slavery, it has been impossible, thus far, to devise a meaningful index of the effect of slavery on the personality or psychology of blacks.

In other words, cliometrics provides a set of tools which are of considerable help in analyzing an important but limited set of problems. Many vital issues in the economics of slavery lie beyond the range of these tools. This does not mean that issues such as the quality of slave family life or the psychological effects of slavery on black personalities should be ignored, but that discussion of these issues must of necessity continue to be shrouded in more uncertainty than discussion of such questions as the nutritional ade-

quacy of the slave diet, the profitability of investments in slaves, and the efficiency of slave labor.

This book not only presents the principal findings of the cliometricians but also attempts to interpret them. In the pages that follow, we reconsider many aspects of the prevailing interpretation of the slave experience and of black history in light of the new findings. It is important that readers keep in mind this distinction between the principal findings of the cliometricians and our attempt to interpret them. The findings and the interpretation do not stand on the same level of certainty. Interpretation sometimes involves additional data which are quite fragmentary and assumptions which, though they are plausible, cannot be verified at present. Hence, even when readers accept the validity of one or another of the principal findings, they may disagree with the significance that we attach to it.

There is no such thing as errorless data. The questions with which cliometricians must always grapple are the nature of the errors contained in the various bodies of data that they employ and the biases that such errors will produce in estimates based on data containing them. When evaluated from this standpoint, evidence falls not into just two categories (good and bad), but into a complex hierarchy in which there are many categories of evidence and varying degrees of reliability.

Our aim has been to base our statements on evidence drawn from as high up in this hierarchy of reliability as possible. We have least confidence in fragmentary evidence which is based on unverifiable impressions of individuals whose primary aim was the defense of an ideological position. Fragmentary evidence from objective sources, such as impressionistic reports of "detached" observers, may be more believable, but is still of a low order of reliability since it is usually not possible to submit such evidence to systematic statistical tests.

Data which can be subjected to systematic statistical tests

(which we will refer to as "systematic data"), even when they are biased, are frequently more useful than fragmentary data. For if the nature or direction of the bias is known, it is frequently possible to design statistical procedures which will yield valid tests of important historical issues. For example, in chapter 4 we present an estimate of the nutritional content of the slave diet on large plantations. The first step in the estimation procedure involved the determination of the total supply of food available for human consumption on the designated plantations. The second step required the separation of the consumption of slaves from that of the whites who lived on the plantations. Since no systematic information on the diets of these whites was available, we resolved the issue by assuming that their consumption of particular foods was substantially in excess of national averages. This "overfeeding" resulted in an estimated daily intake for whites of 5,300 calories — a figure which is clearly in excess of any feasible level of consumption for the relatively inactive, upper-class whites living on plantations. While the "overfeeding" of whites leads to an underestimation of slave consumption, the error is relatively small (less than 5 percent of the per capita consumption of slaves), since whites formed a small percentage of the population of large plantations. Moreover, correction of this small error would not change the conclusion that the nutritional content of the slave diet was quite high. Quite the contrary, correction of the error would only serve to buttress that conclusion.

We use fragmentary, impressionistic evidence in two ways: to illustrate and make more vivid results that have been established by more precise methods, and to fill in gaps in evidence where it has not been possible thus far to obtain systematic data. Arguments that rest on impressionistic, fragmentary evidence must be considered to be on a relatively low level of reliability, regardless of the objectivity of the source of this evidence.

In order to encourage the widest possible discussion of the findings of the cliometricians, we have attempted to present them in as nontechnical a manner as possible. Popularization has its costs. Those who are familiar with cliometric writings know that one of its principal hallmarks is "sensitivity analysis." "Sensitivity analysis" refers to the investigation of the quantitative effects on findings of possible biases in the data, in the behavioral equations, and in the assumptions of the estimating techniques. However, if we attempted to introduce such analysis into the main body of the text, the book would quickly become too technical.

Instead we have included a technical appendix (appendix B) which discusses the data on which the principal findings are based, as well as the techniques employed in the analyses of these data. While not all of the technical issues are included in appendix B, we deal there with the most important issues, or at least with the most controversial ones. Those readers who desire more complete information should consult the various scientific papers on which this book is based. Papers that have already been published or that will soon be published are referred to in appendices B or C and are cited in the list of references.

Appendixes B and C and the list of references, as well as an appendix on the role of science in the study of history, are to be found in the companion volume to *Time on the Cross*, which is subtitled *Evidence and Methods: A Supplement*.

One.

The International Context of United States Slavery

The Origins of the Atlantic Slave Trade

Slavery is not only one of the most ancient but also one of the most long-lived forms of economic and social organization. It came into being at the dawn of civilization, when mankind passed from hunting and nomadic pastoral life into primitive agriculture. And although legally sanctioned slavery was outlawed in its last bastion — the Arabian peninsula — in 1962, slavery is still practiced covertly in parts of Asia, Africa, and South America.

Over the ages the incidence of slavery has waxed and waned. One high-water mark was reached during the first two centuries of the Roman Empire when, according to some estimates, three out of every four residents of the Italian peninsula — twenty-one million people — lived in bondage. Eventually Roman slavery was transformed into serfdom, a form of servitude that mitigated some of the harsher features of the older system. Serfs were perpetually tied to land and owed a certain amount of labor service to

prices, for example, frequently rise with political tensions as investors contemplate the possible effects of a war on profits. The indexes portrayed in figure 32 suggest that in the late 1850s slaveholders expected the political policies pursued by their leaders, including the possibility of war, to lead to an improvement of their economic position. The South, of course, had its Cassandras. But during 1855-60, slaveholders as a group did not anticipate the complete collapse of their peculiar institution that was to follow from the Civil War.

Four.

The Anatomy of Exploitation

Webster's Third New International Dictionary gives two definitions of personal exploitation. These are:

1. an unjust or improper use of another person for one's own profit or advantage;
2. utilization of the labor power of another person without giving a just or equivalent return.

Slaves were exploited in both of these senses. For the advantage of their masters, they were whipped, sold on the auction block, separated from loved ones, deprived of education, terrorized, raped, forced into prostitution, and worked beyond limits of human endurance. The labor power of slaves was also utilized without giving slaves an equivalent return. This aspect of their exploitation was most apparent in the case of hires. The man who rented a slave paid the full market value of the slave's services, but the slave received only part of that payment. The slave's income was the expenditure of the renter on his maintenance; the balance of the value produced by the slave went to the owner of the slave in the form of a rental fee.

While the existence of exploitation is beyond question, the extent of that exploitation is less clear. How frequent was the mean and improper use of slaves and how far did meanness go? How much of the income produced by slaves was expropriated from them?

The posing of these questions may seem irrelevant, even malicious. For many, it is enough merely to recognize the existence of 250 years of exploitation under slavery and to stress its horror. To haggle over the extent of the exploitation suggests callousness to the agony of human bondage, and appears to diminish the significance of the moral issue.

If all that was at stake was the refinement of the historical image of the slaveholding class, the issue of the extent of exploitation could be disregarded. Slaveholders have long been driven from the stage of American life, and the descendants of slaveholders generally do not wish to dwell on the meaner aspects of their ancestry. It is the need to arrive at an accurate historical image of the black man that gives urgency to the issue.

For it is widely assumed that the plantation regime under which most slaves lived was so cruel, the exploitation so severe, the repression so complete, that blacks were thoroughly demoralized by it. In this view, blacks were virtually cultural ciphers until they obtained their freedom in 1865. Little positive development of black culture or personality was possible under the unbridled exploitation of slavery; those developments which did take place were largely negative. Under the regime of the lash the blacks had little incentive to improve themselves and "got into very bad habits of doing as little as possible." Undernourished, if not starved, they lacked both the physical capacity and the mental stamina to tackle any but the most routine tasks, and even these were poorly carried out. Forced or encouraged into promiscuity, blacks became extremely casual in their sexual relations: sexual rivalries threw "many slave quarters in constant turmoil"; "[c]hasity was 'out of the question'."

and many girls became pregnant at twelve, thirteen, and fourteen years of age. Sexual laxity on the part of the slaves, combined with a wide array of policies pursued by masters, reduced the black family to "cultural chaos." Bereft of "deep and enduring affection," fathers and mothers not only came to regard their children with "indifference" but often neglected them in sickness and even practiced infanticide.

No one's personality could, according to this view, fail to be affected by a regime so brutal that some have compared plantations to concentration camps and others have compared them to prisons. In the daily fire of such "total" exploitation, masters and overseers fashioned a distinctive type of "slavish personality" that Stanley M. Elkins identified as "Sambo."

Sambo, the typical plantation slave, was docile but irresponsible, loyal but lazy, humble but chronically given to lying and stealing; his behavior was full of infantile silliness and his talk was inflated by childish exaggeration. His relationship with his master was one of utter dependence and childlike attachment; it was indeed this childlike quality that was the very key to his being.

This, then, is the portrait contained in many current histories of the antebellum South. Both masters and slaves are painted as degraded brutes. Masters are vile because they are the perpetrators of unbridled exploitation; slaves are vile because they are the victims of it. How true to life is the portrait?

Food, Shelter, and Clothing

The belief that the typical slave was poorly fed is without foundation in fact. This mistaken view may have arisen from

a misinterpretation of the instructions of masters to their overseers. For these documents often mention only corn and pork in outlining the rations that were to be distributed to Negroes. The typical daily ration described was two pounds of corn and one half pound of pork per adult. The misinterpretation stems from the incorrect assumption that the lack of reference to other foods meant that the slave diet was restricted largely to corn and pork.

Overseers' instructions, however, were not book-length manuals. They usually occupied just a few handwritten pages (generally ranging between one thousand and three thousand words in total) and were confined to outlining the major features of the routine the master wished to be pursued. They were not meant to be exhaustive documents, but to underscore those aspects of plantation management that particular owners held to be especially important. The incomplete nature of these widely quoted instructions is seen by the frequent omission of such important matters as rations for children, the disposition of wheat and other grain crops besides corn, the care of chickens and other small livestock, the production and disposition of dairy products, and the different feeds to be used for the various categories of livestock. The instructions to overseers are useful not because they contain a complete description of the plantation routine but because they reveal which aspects of that routine were uppermost in the minds of plantation owners.

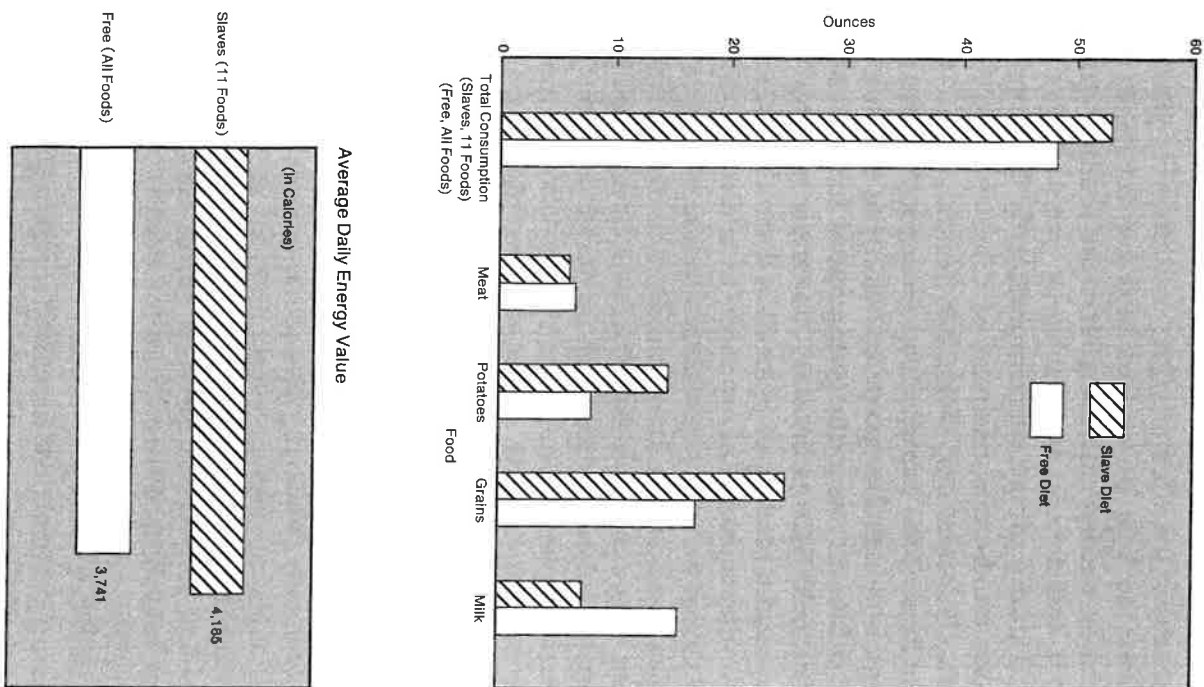
The rationing of corn and pork to slaves was emphasized by slaveholders for two reasons. First, while corn and pork did not constitute the totality of the slave diet, they were the core of the diet on most plantations. Unlike other foods such as fruit and vegetables which were fed to slaves only in certain seasons, daily, weekly, or monthly rations of corn and pork were distributed throughout the year. Secondly, while beef, chickens, dairy products, and Irish potatoes had to be consumed soon after they were

slaughtered or harvested (because they were difficult to preserve for later use), pork and corn were kept in store for the full year. Winter, especially January and February, was the season for killing, curing, and smoking pork, which could then be kept under lock and key in smoke houses until it was needed. Ensuring that the store of pork and corn was sufficient to last the entire year was one of the principal duties of the overseer, which is why explicit instructions on the disbursement of these two staples were common. The overseer, wrote one master, "must himself keep the keys of the [corn] cribs, smokehouses and all other buildings in which any property belonging to me is stored and must himself see to the giving out of food." Feed everyone "plentifully," wrote another, "but waste nothing."

More careful reading of plantation documents shows that the slave diet included many foods in addition to corn and pork. Among the other plantation products which slaves consumed were beef, mutton, chickens, milk, turnips, peas, squashes, sweet potatoes, apples, plums, oranges, pumpkins, and peaches. Certain foods not produced on most plantations were frequently purchased for slave consumption, including salt, sugar, and molasses. Less frequent, but not uncommon, purchases for slaves included fish, coffee, and whiskey. In addition to food distributed to them, slaves supplemented their diet, in varying degrees, through hunting and fishing as well as with vegetables grown in the garden plots assigned to them.

Unfortunately, surviving plantation records are not complete enough to permit determination of the average amounts of each of the foods purchased for slaves or of the quantities of meats and fish that slaves obtained through hunting and fishing. However, on the basis of data obtained from the manuscript schedules of the 1860 census, it has been possible to compute the average amounts of eleven of the principal foods consumed by slaves who lived on the large plantations of the cotton belt. These eleven foods are

A Comparison of the Average Daily Food Consumption of Slaves in 1860 with the Average Daily Food Consumption of the Entire Population in 1879



beef, pork, mutton, milk, butter, sweet potatoes, white potatoes, peas, corn, wheat, and minor grains. While this list is short, it probably accounts for 80 percent of the caloric intake of slaves. Fish, fowl, game, sugar, as well as the various omitted vegetables and fruits, were choice items; but they did not constitute a large part of the diet for either whites or blacks during the middle of the nineteenth century.

Figure 33 shows that the average daily diet of slaves was quite substantial. The energy value of their diet exceeded that of free men in 1879 by more than 10 percent. There was no deficiency in the amount of meat allotted to slaves. On average, they consumed six ounces of meat per day, just an ounce lower than the average quantity of meat consumed by the free population. While pork was more important in the slave than in the free diet, the difference was not as large as is usually presumed. Slaves averaged 70 percent of the free population's consumption of beef. The milk consumption was low by free standards, but still amounted to about one glass per day for each slave.

By weight, grains and potatoes dominated the diet of both the free and slave population. Much has been made of the fact that corn was the principal grain consumed by slaves, while wheat was the principal grain in the free diet. Yet from a nutritional standpoint, both are excellent foods, high in energy value and with substantial protein content. Wheat is richer in calcium and iron, but corn has more vitamin A. What has completely escaped attention is the fact that while both slaves and free men ate large quantities of potatoes, slaves consumed virtually nothing but sweet potatoes, although most of the potatoes consumed by free men were white. The significance of this dichotomy is that sweet potatoes are a much better food than white potatoes. Sweet potatoes are especially rich in vitamins A and C and are also fairly high in calcium.

The high slave consumption of meat, sweet potatoes, and peas goes a long way toward explaining the astounding

Figure 34

The Nutritional Value of the Slave Diet: Average Slave Consumption of Various Nutrients in 1860 as a Percentage of Modern Recommended Daily Allowances

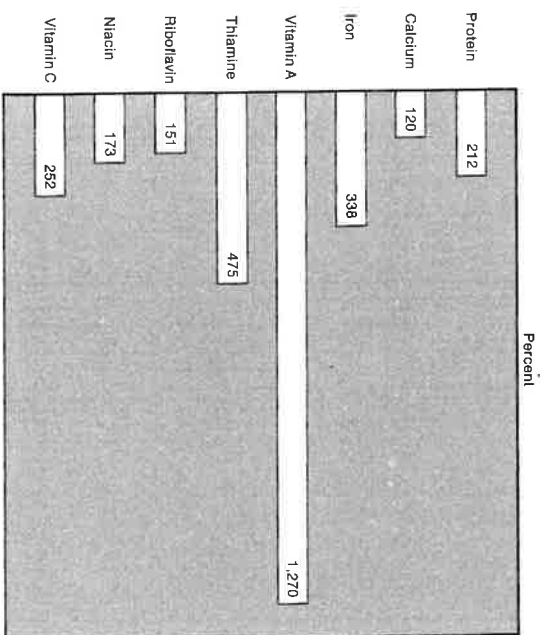
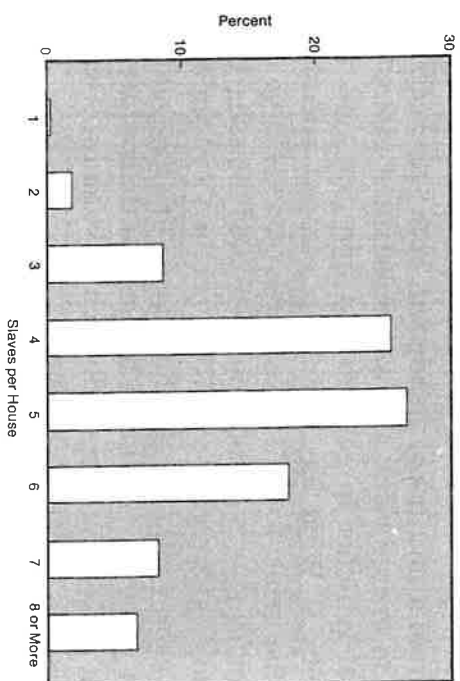


Figure 35

The Distribution of Slaves on Large Plantations, by Persons per Slave House, 1860



results shown in figure 34. The slave diet was not only adequate, it actually exceeded modern (1964) recommended daily levels of the chief nutrients. On average, slaves exceeded the daily recommended level of proteins by 110 percent, calcium by 20 percent, and iron by 230 percent. Surprisingly, despite the absence of citrus fruits, slaves consumed two and one half times the recommended level of vitamin C. Indeed, because of the large consumption of sweet potatoes, their intake of vitamin A was at the therapeutic level and vitamin C was almost at that level. Of course, the fact that the *average* daily nutrient content of the slave diet was good does not mean that it was good for all slaves. And even the best-fed slaves experienced seasonal variation in the quality of their diet, due to the limitations in the technology of food preservation during the antebellum era.

Data on slave housing are much more sparse than on slave diets. The most systematic housing information comes from the census of 1860, which included a count of slave houses. These census data show that on average there were 5.2 slaves per house on large plantations. The number of persons per free household in 1860 was 5.3. Thus, like free men, most slaves lived in single-family households. The sharing of houses by several families of slaves was uncommon. Occasionally, on very large plantations, there were dormitories for unmarried men and women. But these were exceptional. The single-family household was the rule.

Unfortunately, the census did not collect information on the size or the quality of slave houses. Descriptions in plantation records and in travelers' accounts are fragmentary. They suggest a considerable range in the quality of housing. The best were three- or four-room cottages, of wood frame, brick, or stone construction, with up to eight hundred square feet of space on the inside, and large porches on the outside. Such cottages had brick or stone chimneys and glazed windows. At the other pole were single-room log cabins

without windows. Chimneys were constructed of twigs and clay; floors were either earthen or made of planks resting directly on the earth.

Comments of observers suggest that the most typical slave houses of the late antebellum period were cabins about eighteen by twenty feet. They usually had one or two rooms. Lofts, on which the children slept, were also quite common. Windows were not glazed, but closed by wooden shutters. Some houses also had rear doors. Chimneys were usually constructed of brick or stone. The building material was usually logs or wood. Seams in the log cabins were sealed by wooden splints and mud. Floors were usually planked and raised off the ground.

While such housing is quite mean by modern standards, the houses of slaves compared well with the housing of free workers in the antebellum era. It must be remembered that much of rural America still lived in log cabins in the 1850s. And urban workers lived in crowded, filthy tenements. One should not be misled by the relatively spacious accommodations in which U.S. working-class families live today. That is an achievement of very recent times. As late as 1893, a survey of the housing of workers in New York City revealed that the median number of square feet of sleeping space per person was just thirty-five. In other words, the "typical" slave cabin of the late antebellum era probably contained more sleeping space per person than was available to most of New York City's workers half a century later.

The best information on clothing comes from the records of large plantations. These indicate that a fairly standard annual issue for adult males was four shirts (of cotton), four pairs of pants (two of cotton and two of wool), and one or two pairs of shoes. Adult women were issued four dresses per year, or the material needed to make four dresses. Hats were also typically issued annually (women received headkerchiefs). Blankets were issued once every two or three years. There seems to have been much more variability in

the issue of socks and underclothes. Issues of petticoats to women are mentioned in a few records, but not in most. Mention of socks and underwear for men is also irregular. For winter months men had jackets, sometimes overcoats, although the frequency of their issue is unclear. Clothing for children showed some variation from estate to estate, but by far the most common issue was a one-piece garment which looked like an extra-long shirt.

Slave clothing was usually made of a coarse but durable cloth. The leather in slave shoes was of a high grade, but little attention was devoted to matters of fashion. Finer clothes were supplied to house servants and other favored slaves. Slaves also supplemented the standard issue by their own purchases. As indicated below, many slaves were able to earn substantial sums of money. Much of this was spent on such items of clothing as headkerchiefs and brightly colored cloth for dressmaking.

Medical Care

While the quality of slave medical care was poor by modern standards, there is no evidence of exploitation in the medical care typically provided for plantation slaves. The inadequacy of the care arose not from intent or lack of effort on the part of masters, but from the primitive nature of medical knowledge and practices in the antebellum era.

That adequate maintenance of the health of their slaves was a central objective of most planters is repeatedly emphasized in instructions to overseers and in other records and correspondence of planters. "The preservation of the health of the negroes," wrote J. A. S. Acklen to his overseer, "and the care of them when sick, will require your best attention; and to be ignorant of the best mode of discharg-

ing your duties in these particulars, is to be unfit for the responsible station you hold." P. C. Weston charged his

Table 2

Clothing Recommended for an Adult Worker by Social-work Agencies in New York City in 1907 Compared with the "Typical" Clothing Allotment for an Adult Male Slave

NEW YORK WORKER	"TYPICAL" SLAVE ISSUE
2 hats or caps	1 hat
1 overcoat (every 2 or 3 years)	? overcoat
1 suit	1 jacket
1 pair pantaloons	4 pair pants
2 pair overalls	
5 shirts	4 shirts
6 collars	
4 ties	
4 handkerchiefs	
summer and winter underwear	? underwear
6 pair hose	? hose
2 pair shoes	1½ pair shoes
repair of shoes	
gloves or mittens	

The "typical" slave issue does not include purchases of clothing by slaves. The New York study in which the above list was published reported that the average expenditure of New York workers on clothes was below the recommended level.

overseer "most distinctly to understand that his first object is to be, under all circumstances, the care and well being of the negroes." Consequently,

All sick persons are to stay in the hospital night and day, from the time they first complain to the time they are able to work again. The nurses are to be responsible for the sick not leaving the house, and for the cleanliness of the bedding, utensils, &c. The nurses are never to be allowed to give any medicine without the orders of the Overseer or Doctor. A woman, beside the plantation nurse, must be put to nurse all persons seriously ill. In all cases at all serious the Doctor is to be sent for, and his orders are to be strictly attended to; no alteration is to be made in the treatment he directs.

While planters worried about slaves who feigned illness to get out of work, they were generally more concerned about losing slaves or impairing their health through the neglect of real illness. Thus Bennet H. Barrow, owner of one of the largest Louisiana plantations, treated slaves as though they were sick even when he thought they were pretending. Nor was Barrow alone in this attitude. In their sick records planters sometimes described the malady which had removed a slave from production as "nothing," "complaining," and "more lazy and mad than sick." James Hammond insisted that, "[e]very reasonable complaint must be promptly attended to; and with any marked or general symptom of sickness, however trivial, a negro may lie up a day or so at least." "Unless it is a clear case of imposition," wrote still another planter, "a negro had better be allowed a day's rest when he lays up. A little rest often saves much by preventing serious illness."

Facilities for the treatment of the sick generally varied with the size of the plantations. The larger plantations maintained substantial hospitals. On one plantation with 168 slaves, for example, the hospital was a two-story brick building which had eight large rooms. Such hospitals usually contained separate rooms for men and women. A special room was often set aside for confinement cases. One or more of the rooms were "clinics" for the treatment of "outpatients." These rooms contained a pharmacy as well as other equip-

ment needed for ministering to the sick. On smaller plantations, the "hospital" was merely an ordinary cabin reserved for the sick. In some instances, masters set aside several rooms in their own houses for use as a hospital. The rationale for hospitalizing slaves was twofold: it permitted the sick to receive special care including not only rest and medication but also special diets; it also isolated the sick slaves from the healthy ones and thus minimized the danger of contagion. Many planters insisted that slaves be removed from their cabins to hospitals as soon as their illness was made known.

Few plantations were large enough to justify the exclusive retention of a full-time physician. However, virtually all plantations of moderate or large size had at least one full-time nurse, usually an elderly slave, and many also had experienced midwives. The nurses and midwives worked under the direct supervision of the planter or his overseer. Planters sought to be, and overseers were expected to be, knowledgeable about current medical procedures and about drugs and their administration. Physicians were regularly brought onto the plantations to care for slaves whose health problems could not be treated adequately by the nurses, midwives, overseers, or planters. Some planters contracted for the physicians' services for a year at a time, paying a flat fee that was usually proportional to the number of persons covered by the contract. Others paid for services as rendered. In either case, the doctor attending to the slaves was usually the same doctor who ministered to the planter's family. Bills submitted by these physicians indicate, time after time, that they treated both slaves and members of the master's family during the same visit.

That it was generally the intent of planters to supply slaves with medical care of a relatively high quality does not imply that the objective was usually realized. Not only was the state of medical knowledge and arts quite primitive during the antebellum era, but the prevailing theory of

disease frequently led to treatments which were inimical to the recovery of patients. Disease was assumed to be caused by "poisons emanating from decaying animal and vegetable matter," which were transmitted to human beings by "impure airs and waters." To rid the body of these poisons, doctors resorted to bleeding, blistering, and purging. Bleeding and purging were a standard treatment for such diseases as dysentery, cholera, and pleurisy. In each case, of course, the prescribed treatment removed from the body fluids that were vital to recovery. The therapy, no doubt, hurried to their graves many patients who might have survived if they were spared the services of doctors.

For many of the illnesses that afflicted slaves, especially pneumonia and diseases of the gastrointestinal tract, which were the greatest killers of blacks, the services of doctors were either useless or harmful. It was only in limited cases, such as smallpox and malaria (after the 1820s when sulphate of quinine became generally known), that doctors possessed effective pharmaceutical weapons. Beyond these, the useful medicines contained in the chests of doctors were primarily anodynes such as opium, paregoric, Dover's powder, and laudanum which, although they did not cure, at least served to relieve severe distress. Doctors were effective in cases requiring minor surgery, such as opening abscesses, removing teeth, and the setting of broken bones. They were also helpful in dealing with hernias, which were quite common among slaves. Although they lacked the surgical knowledge needed to repair the rupture, they did prescribe the use of trusses.

The prevailing theory of disease did have one quite salutary effect. It led many planters to adhere to a hygienic regimen. Few matters were more frequently emphasized in the instructions to overseers than the need to insure not only the personal cleanliness of slaves but also the cleanliness of their clothes, their bedding, and their cabins. Instructions required overseers "personally" to

see that they keep their clothing mended and clean, and that they wash their clothes as often as once a week, for which purpose time must be regularly set apart the latter end of the week. He must see that they are clean on Sundays and not straggling about the country dirty and ragged, and he must see that they appear clean every Monday morning in the year, without any failure whatever.

He must see that they keep their houses clean and their yards free from weeds and filth.

Some planters went even further. Hammond required a complete cleaning of slave cabins twice a year. During these spring and fall cleanings, all contents of the cabin were to be removed and sunned, all walls were to be washed, mattresses were to be restuffed, and the grounds under the houses were to be sprinkled with lime. Once a year every house was to be "whitewashed inside and out." The punishment for slaves who failed to keep themselves personally clean was a forced scrubbing by the driver and two other blacks. Another planter required the ground under the cabins to be swept every month and had cabins whitewashed twice a year. Charles Tait, a leading planter in Alabama, required the cabins of his slaves to be moved to new ground every fourth year "to prevent filth accumulations and cholera or diphtheria."

Slave health care was at its best for pregnant women. "Pregnant women," wrote one planter, "must be treated with great tenderness, worked near home and lightly." "Light work" was generally interpreted as 50 to 60 percent of normal effort and was to exclude activity which required heavy physical effort. During the last month of pregnancy work was further reduced, although various planters felt, as did P. C. Weston, that "pregnant women are always to do some work up to the time of their confinement, if it is only walking into the field and staying there." On large estates, women were usually confined in the "lying-in" ward of the plantation hospital. In normal cases the delivery was

usually handled by a midwife. But if some difficulty in the delivery was anticipated, a doctor was called in. The period of confinement generally lasted about four weeks, during part of which time mothers were attended to by a midwife or a nurse. There frequently followed another two weeks of light work in the vicinity of the slave quarters. Women were expected to nurse their children, and hence were kept on relatively light work schedules for the balance of the year. Until the sixth or eighth month after birth, nursing took place four times per day during working hours, and nursing mothers were expected to work at only 50 to 60 percent of normal levels. For the balance of the first year infants were usually nursed twice a day during hours of work.

Demographic evidence gives strong support to descriptions of pre- and post-natal care contained in plantation rules, letters, and diaries. Computations based on data from the 1850 census indicate that the average death rate due to pregnancy among slave women in the prime childbearing ages, twenty to twenty-nine, was just one per thousand. This means that out of every 167 women in this age category who gave birth, only one died. The slave mortality rate in childbearing was not only low on an absolute scale, it was also lower than the maternal death rate experienced by southern white women.

The mortality rate for infants was less favorable than that of their mothers. Of every thousand slaves born in 1850, an average of 183 died before their first birthday. The death rate for white infants in the same year was 146 per thousand. In other words, the infant death rate for slaves was 25 percent higher than for whites. This finding appears to give credence to charges that mean treatment of infant slaves was widespread. In so doing it raises the strange paradox of planters who treated pregnant women and new mothers quite well while abusing their offspring.

Work on the demographic data for 1850 and 1860 has not

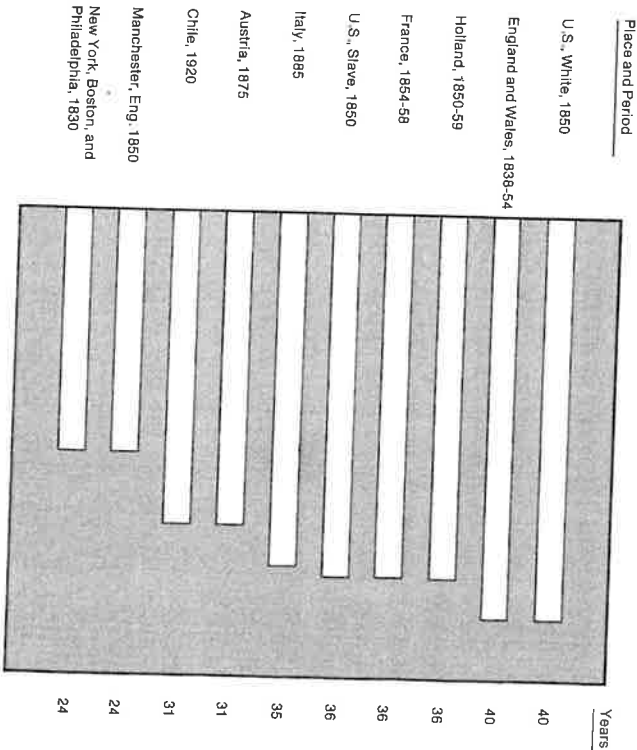
proceeded far enough to permit a full resolution of this paradox. Definitive answers to many of the questions posed by the relatively high infant death rates experienced by slaves are at least several years off. The work thus far, however, has revealed little evidence to support the charge that masters neglected the care of infants. Most of the difference between the infant death rates of slaves and free persons appears to have been due to the fact that the South was less healthy than the North. The infant death rate of southern whites in 1850 was 177 per thousand — virtually the same as the infant death rate of slaves. Along this dimension, then, exploitation of blacks arose because slavery was confined to the South and because slaves were not free to choose where to live.

By and large, slave deaths during the first year of life, like those of free infants, were due to diseases such as whooping cough, croup, pneumonia, cholera, and various maladies of the gastrointestinal system — diseases about which the men of the antebellum era had little understanding and over which they had little control. Of those causes of death that apparently could be controlled, suffocation was most significant. About 9.3 percent of the slave infants who died in 1850 were reported to have succumbed from this cause. Among whites, only 1.2 percent of infants were reported to have died from suffocation. The excess of the slave suffocation rate over the white suffocation rate accounted for nearly fifteen deaths out of every thousand slave births.

There has been much debate among historians as to the causes of infant suffocations. Some have attributed them to infanticide, arguing that life was so unbearable for slaves that many mothers preferred to kill their young rather than to rear them in bondage. This argument is undermined by the extremely low suicide rate among slaves. Less than one slave in every ten thousand committed suicide in 1850. That was only one third of the suicide rate among the

Figure 36

The Life Expectation at Birth for U.S. Slaves and Various Free Populations, 1830-1920



white population. Others have used suffocation as evidence that neglect of children by slave mothers was widespread. While one cannot rule out the possibility that deaths due to suffocation reflected neglect on the part of the mothers involved, the fact remains that out of every one hundred slaves born, less than two infants died of this cause. The mortality statistics, therefore, leave plenty of room for slave mothers who gave their children tender care.

Nor should one necessarily accept the statistics on slave suffocations at face value. Many deaths that were interpreted as suffocation because of the absence of a prior fever

or other then-accepted symptoms of illness could have been caused by undisclosed infections. That more such deaths were reported as suffocation for slaves than for free men might have been due to the jaundiced view of the overseers who reported the death statistics to the census takers.

Neither the 1850 nor the 1860 censuses collected information on morbidity. The only currently available data comes from a sample of 545 field hands who lived on fifteen plantations. This sample provides information on illness rates during 2,274 man years of labor time. It shows that on average each slave was sufficiently ill to be absent from work for just 12.0 days per year. This low illness rate tends to support the impression that slaves were well cared for. It also calls into question the widely asserted contention that slaves were always feigning illness. At the very least, the low absence rate shows that if pretense of illness was widespread, slaves were singularly unsuccessful in their deceptions.

For many, statistics on life expectancy are the ultimate measure of physical well-being. Figure 36 compares the life expectancy of U.S. slaves in 1850 with those of free men in various places between 1830 and 1920. Although the life expectation of slaves in 1850 was 12 percent below the average of white Americans, it was well within the range experienced by free men during the nineteenth century. It was, for example, nearly identical with the life expectation of countries as advanced as France and Holland. Moreover, U.S. slaves had much longer life expectations than free urban industrial workers in both the United States and Europe.

The Family

The administration of most large plantations was based on two organizations. Fieldwork revolved around gangs.

They were the vehicles through which planters were able to achieve a degree of specialization and efficiency that was unmatched elsewhere in agriculture. This aspect of plantation organization is considered in chapter 6.

The other organization of central importance was the family. Planters assigned three functions to the slave family. First, it was the administrative unit for the distribution of food and clothing and for the provision of shelter. As we have already seen, the single-family house, not the dormitory, was almost the universal form of shelter on larger plantations. The records of planters also indicate that whether food was cooked in a common kitchen or in the house of individual families, allotments of rations were generally made by family. The same was true for clothing.

The family was also an important instrument for maintaining labor discipline. By encouraging strong family attachments, slaveowners reduced the danger that individual slaves would run away. By permitting families to have *de facto* ownership of houses, furniture, clothing, garden plots, and small livestock, planters created an economic stake for slaves in the system. Moreover, the size of the stake was variable. It was possible for some families to achieve substantially higher levels of income and of *de facto* wealth ownership than others. The size and quality of houses and the allotments of clothes as well as the size of the garden plots differed from family to family.

Third, the family was also the main instrument for promoting the increase of the slave population. Not only did planters believe that fertility rates would be highest when the family was strongest, but they relied on the family for the rearing of children. Although infants and very young children were kept in nurseries while mothers labored in the field, these supplemented rather than replaced the family. The central importance of the family in the rearing of slaves is revealed by the narratives of former slaves collected by the

W.P.A. in the 1930s. In discussing their early upbringing and the influences on them, the former slaves frequently refer to what their parents taught them but rarely, if ever, invoke the names of women who ran the nurseries.

To promote the stability of slave families, planters often combined exhortations with a system of rewards and sanctions. The rewards included such subsidies as separate houses for married couples, gifts of household goods, and cash bonuses. They often sought to make the marriage a solemn event by embedding it in a well-defined ritual. Some marriage ceremonies were performed in churches, others by the planter in the "big house." In either case, marriages were often accompanied by feasts and sometimes made the occasion for a general holiday. The sanctions were directed against adultery and divorce. For many planters, adultery was an offense which required whippings for the guilty parties. Some planters also used the threat of the whip to discourage divorce.

Thus, while the existence of slave marriages was explicitly denied under the legal codes of the states, they were not only recognized but actively promoted under plantation codes. That the legal basis for slave marriage was derived from codes which held sway within the jurisdiction of the plantation, points to a much neglected feature of legal structure of the antebellum South. Within fairly wide limits the state, in effect, turned the definition of the codes of legal behavior of slaves, and of the punishment for infractions of these codes, over to planters. Such duality of the legal structure was not unique to the antebellum South. It existed in medieval Europe in the duality between the law of the manor and of the crown; it was a characteristic of the regimes under which the American colonies were governed; and in lesser degree, it exists with respect to certain large institutions today (for example, with respect to university regulations).

The importance of the dual legal structure of the ante-

bellum South is that the latitude which the state yielded to the planter was quite wide. For most slaves it was the law of the plantation, not of the state, that was relevant. Only a small proportion of the slaves ever had to deal with the law-enforcement mechanism of the state. Their daily lives were governed by plantation law. Consequently, the emphasis put on the sanctity of the slave family by many planters, and the legal status given to the slave family under plantation law, cannot be lightly dismissed.

Recognition of the dualistic nature of the southern legal structure puts into a different perspective the emphasis which some historians have placed on the "pre-bourgeois" character of antebellum society. What made that society pre-bourgeois was not the absence of a commercial spirit among planters, but the wide area of legal authority which the state yielded to them. In Europe, the rise of capitalism was accompanied by a determined struggle to weaken the authority of the manor and to transfer its powers to the centralized state. This process was sharply curtailed in the South. While the South developed a highly capitalistic form of agriculture, and while its economic behavior was as strongly ruled by profit maximization as that of the North, the relationship between its ruling and its servile class was marked by patriarchal features which were strongly reminiscent of medieval life. Unlike the northern manufacturer, the authority of the planter extended not only to the conduct of business but to the regulation of the family lives of slaves, the control of their public behavior, the provision of their food and shelter, the care of their health, and the protection of their souls.

We do not mean to suggest that planters viewed the slave family purely as a business instrument. Victorian attitudes predominated in the planting class. The emphasis on strong, stable families, and on the limitation of sexual activity to the family, followed naturally from such attitudes. That morality and good business practice should coincide

created neither surprise nor consternation among most planters.

Of course not all planters, and not all of their overseers, were men who lived by the moral codes of their day. That many of these men sought sex outside of the confines of their wives' beds is beyond question. To satisfy their desires they took on mistresses and concubines, seduced girls of tender ages, and patronized prostitutes. Such sexual exploitation was not limited to the South. And within the South, sexual exploitation by white men was not limited to black women.

The point at issue here is not whether the sexual exploitation of slave women by masters and overseers existed, but whether it was so frequent that it undermined or destroyed the black family. Let us pose the question somewhat more sharply: Are there reasons to believe that the degree of sexual exploitation which white men imposed on black women was greater than that imposed on white women? We put the issue in this way because while the sexual exploitation of white women was rife, few have gone so far as to claim that such exploitation destroyed the family institution among whites. Is the asymmetry in the presumed effects of sexual exploitation on the families of blacks and whites justified by available evidence?

Antebellum critics of slavery answered these questions in the affirmative. They accused slaveowners and overseers of turning plantations into personal harems. They assumed that because the law permitted slaveowners to ravish black women, the practice must have been extremely common. They also assumed that black women were, if not more licentious, at least more promiscuous than white women, and hence less likely to resist sexual advances by men, whether black or white. Moreover, the ravishing of black women by white men was not the only aspect of sexual exploitation which devastated the slave family. There was also the policy of deliberate slave-breeding, under which

planters encouraged promiscuous relationships among blacks. Thus, economic greed and lust on the part of the planters, and submissiveness on the part of the slaves, combined to make the sexual exploitation of black women so extreme as to be beyond comparison with the situation of white women.

The evidence on which these assumptions and conclusions were based was extremely limited. While none of the various travelers through the South had seen deliberate slave-breeding practiced, they had all heard reports of it. Some travelers published conversations with men who admitted to fathering a large number of the slaves on their plantations. Others wrote of the special solicitude shown by one or another master to mulatto offspring, a solicitude which in their minds strongly implied parenthood. There were also the descriptions of the treatment of especially pretty slave women on the auction block and of the high prices at which such women sold, prices too high to be warranted by field labor and which could be explained only by their value as concubines or as prostitutes.

Even if all these reports were true, they constituted at most a few hundred cases. By themselves, such a small number of observations out of a population of millions, could just as easily be used as proof of the infrequency of the sexual exploitation of black women as of its frequency. The real question is whether such cases were common events that were rarely reported, or whether they were rare events that were frequently reported. The prevalence of mulattoes convinced not only the northern public of the antebellum era, but historians of today, that for each case of exploitation identified, there were thousands which had escaped discovery. For travelers to the South reported that a large proportion of the slaves were not the deep black of Africans from the Guinea coast but tawny, golden, and white or nearly white. Here was proof beyond denial of either the ubiquity of the exploitation of black women by

white men, or of the promiscuity of black women, or of both.

But this seemingly irrefutable evidence is far from conclusive. It is not the eyesight of these travelers to the South which is questionable, but their statistical sense. For mulattoes were not distributed evenly through the Negro population. They were concentrated in the cities and especially among freedmen. According to the 1860 census, 39 percent of freedmen in southern cities were mulattoes. Among urban slaves the proportion of mulattoes was 20 percent. In other words, one out of every four Negroes living in a southern city was a mulatto. But among rural slaves, who constituted 95 percent of the slave population, only 9.9 percent were mulatto in 1860. For the slave population as a whole, therefore, the proportion of mulattoes was just 10.4 percent in 1860 and 7.7 percent in 1850. Thus it appears that travelers to the South greatly exaggerated the extent of miscegenation because they came into contact with unrepresentative samples of the Negro population. They appear to have had much more contact with the freedmen and slaves of the urban areas than with slaves living in the relative isolation of the countryside. Far from proving that the exploitation of black women was ubiquitous, the available data on mulattoes strongly militates against that contention.

The fact that during the twenty-three decades of contact between slaves and whites which elapsed between 1620 and 1850, only 7.7 percent of the slaves were mulattoes suggests that on average only a very small percentage of the slaves born in any given year were fathered by white men. This inference is not contradicted by the fact that the percentage of mulattoes increased by one third during the last decade of the antebellum era, rising from 7.7 to 10.4 percent. For it must be remembered that mulattoes were the progeny not just of unions between whites and pure blacks but also of unions between mulattoes and blacks. Under common definition, a person with one-eighth ancestry of another race was a mulatto. Consequently, the offspring of two slaves

who were each one-eighth white was to be classified as a mulatto, as was the offspring of any slave, regardless of the ancestry of his or her mate, whose grandfather was a white.

A demographic model of the slave population, which is presented in the technical appendix, shows that the census data on mulattoes alone cannot be used to sustain the contention that a large proportion of slave children must have been fathered by white men. And other available bodies of evidence, such as the W.P.A. survey of former slaves, throw such claims into doubt. Of those in the survey who identified parentage, only 4.5 percent indicated that one of their parents had been white. But the work of geneticists on gene pools has revealed that even the last figure may be too high. Measurements of the admixture of "Caucasian" and "Negro" genes among southern rural blacks today indicate that the share of Negro children fathered by whites on slave plantations probably averaged between 1 and 2 percent.

That these findings seem startling is due in large measure to the widespread assumption that because the law permitted masters to ravish their slave women, they must have exercised that right. As one scholar recently put it, "Almost every [white] mother and wife connected with the institution [of slavery] either actually or potentially shared the males in her family with slave women." The trouble with this view is that it recognizes no forces operating on human behavior other than the force of statute law. Yet many rights permitted by legal statutes and judicial decisions are not widely exercised, because economic and social forces militate against them.

To put the issue somewhat differently, it has been presumed that masters and overseers must have ravished black women frequently because their demand for such sexual pleasures was high and because the cost of satisfying that demand was low. Such arguments overlook the real and potentially large costs that confronted masters and overseers

who sought sexual pleasures in the slave quarters. The seduction of the daughter or wife of a slave could undermine the discipline that planters so assiduously strove to attain. Not only would it stir anger and discontent in the families affected, but it would undermine the air of mystery and distinction on which so much of the authority of large planters rested. Nor was it just a planter's reputation in the slave quarter of his plantation that would be at stake. While he might be able to prevent news of his nocturnal adventure from being broadcast in his own house, it would be more difficult to prevent his slaves from gossiping to slaves on other plantations.

Owners of large plantations who desired illicit sexual relationships were by no means confined to slave quarters in their quest. Those who owned fifty or more slaves were very rich men by the standards of their day. The average annual net income in this class was in excess of \$7,500. That amount was more than sixty times per capita income in 1860. To have a comparable income today, a person would need an after-tax income of about \$240,000 or a before-tax income of about \$600,000. So rich a man could easily have afforded to maintain a mistress in town where his relationship could have been not only more discreet than in the crowded slave quarters of his own plantation, but far less likely to upset the labor discipline on which economic success depended.

For the overseer, the cost of sexual episodes in the slave quarter, once discovered, was often his job. Nor would he find it easy to obtain employment elsewhere as an overseer, since not many masters would be willing to employ as their manager a man who was known to lack self-control on so vital an issue. "Never employ an overseer who will equalize himself with the negro women," wrote Charles Tait to his children. "Besides the morality of it, there are evils too numerous to be now mentioned."

Nor should one underestimate the effect of racism on the

demand of white males for black sexual partners. While some white men might have been tempted by the myth of black sexuality, a myth that may be stronger today than it was in the antebellum South, it is likely that far larger numbers were put off by racist aversions. Data on prostitution supports this conjecture. Nashville is the only southern city for which a count of prostitutes is available. The 1860 census showed that just 4.3 percent of the prostitutes in that city were Negroes, although a fifth of the population of Nashville was Negro. Moreover, all of the Negro prostitutes were free and light-skinned. There were no pure blacks who were prostitutes; nor were any slaves prostitutes. The substantial underrepresentation of Negroes, as well as the complete absence of dark-skinned Negroes, indicates that white men who desired illicit sex had a strong preference for white women.

The failure of Nashville's brothels to employ slave women is of special interest. For it indicates that supply as well as demand considerations served to limit the use of slaves as prostitutes. The census revealed that half of Nashville's prostitutes were illiterate — not functionally illiterate, but completely lacking in either the capacity to read, or to write, or both. In other words, the supply of prostitutes was drawn from poor, uneducated girls who could only command the wages of unskilled labor. Given such a supply, a slaveholder did not have to be imbued with Victorian morals to demur from sending his chattel into prostitution. He could clearly earn more on his slave women by working them in the fields where they would not be subject to the high morbidity and mortality rates which accompany the "world's oldest profession."

The contention that the slave family was undermined by the widespread promiscuity of blacks is as poorly founded as the thesis that masters were uninhibited in their sexual exploitation of slave women. Indeed, virtually no evidence, other than the allegations of white observers, has ever been

presented which sustains the charge that promiscuity among slaves was greater than that found among whites. The question then arises, "Do the allegations reflect the reality of black behavior or are they merely reflections of the preconceptions of the observers?"

The allegations appeared creditable because they emanated not only from southern defenders of slavery but also from critics of the system. While the charges of Southerners could be set down as apologetics, one could not so easily dismiss the words of the abolitionists and other enemies of slavery. On the issue of promiscuity the antislavery forces differed from the apologists not in denying its existence but in the explanation of its extent. Slavery, the critics believed, worked to exacerbate rather than to hold in check the carnal instincts of blacks.

Unfortunately, abolitionists and other antislavery writers were not free of racism merely because they carried the banner of a moral struggle. With their greater physical separation from blacks, these writers were often more gullible and more quick in their acceptance of certain racial stereotypes than slaveholders. This, as we shall see, was a key factor in their underestimation of the efficiency of slave labor. Moreover, coming from the upper classes, as many of these writers did, they shared with slaveholders certain common conceptions regarding the behavior of all laboring folk. Thus, Fanny Kemble, whose descriptions of the family life of slaves is often quoted by historians, saw Irish peasants, English manufacturing workers, and American Negroes as all exhibiting the same "recklessness" toward human propagation.

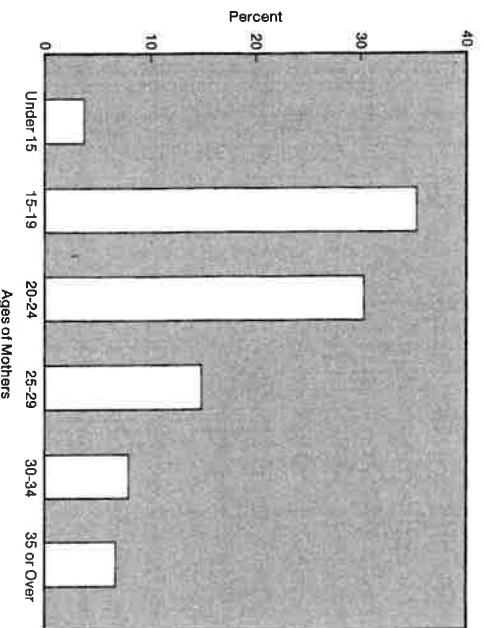
The available demographic evidence on slaves suggests a picture of their sexual lives and family behavior that has little in common with that conveyed by the allegations. One of the most revealing pieces of information is the pattern of child spacing among mothers between the ages of eighteen and thirty. For those whose children were stillborn or

who died within the first three months, the average elapsed time until the birth of the next child was slightly more than one year. However, for those mothers whose children survived the first year of life, the elapsed time before the birth of the next child was somewhat over two years. This is the pattern of child spacing that one would expect to find in a noncontraceptive population in which mothers engaged in breast feeding for the first year of their children's lives. For one of the effects of breast feeding is to reduce the likelihood of conception. In other words, the pattern of child spacing among slaves suggests that the nursing of infants by their mothers was widespread.

This finding hardly supports the charge that slave mothers were indifferent to their children, generally neglected them, and were widely engaged in infanticide. Quite the contrary, the ubiquity of the year-long pattern of breast feeding, combined with the nearly identical rate of infant mortality among slaves and southern whites, and with the rare occurrences of suffocation and other accidents as the cause of death of infant slaves, suggests that for the most part, black mothers cared quite well for their children.

An even more telling piece of information is the distribution of the ages of mothers at the time of the birth of their first surviving child. This distribution, which is shown in figure 37, contradicts the charge that black girls were frequently turned into mothers at such tender ages as twelve, thirteen, and fourteen. Not only was motherhood at age twelve virtually unknown, and motherhood in the early teens quite uncommon, but the average age at first birth was 22.5 (the median age was 20.8). Thus the high fertility rate of slave women was not the consequence of the wanton impregnation of very young unmarried women by either white or black men, but of the frequency of conception after the first birth. By far the great majority of slave children were borne by women who were not only quite mature, but who were already married.

Figure 37

The Distribution of First Births, by the Ages of Slave Mothers

The high average age of mothers at first birth also suggests that slave parents closely guarded their daughters from sexual contact with men. For in a well-fed, noncontraceptive population in which women are quite fecund after marriage, only abstinence would explain the relative shortage of births in the late-teen ages. In other words, the demographic evidence suggests that the prevailing sexual mores of slaves were not promiscuous but prudish — the very reverse of the stereotype published by many in both the abolitionist and slaveholding camps and accepted in traditional historiography. Narratives collected from ex-slaves provide support for the prevalence of prudishness in the conduct of family life. “Dem’s moral times,” recollected Amos Lincoln, who was reared on a plantation in Louisiana. “A gal’s twenty-one fore she marry. They didn’t go wanderin’ round all hours. They

mammies knowed where they was. Folks nowadays is wild and weak.”

That marriage altered the sexual behavior of slaves is clearly indicated by the difference between the seasonal pattern of first births and that of second and subsequent births (see figure 38). Data culled from plantation records indicate that for second and subsequent births, roughly equal percentages of infants were born during every quarter of the year. But the seasonal pattern of first births shows a definite peak during the last quarter of the year — precisely the pattern to be expected in an agrarian society in which a large proportion of marriages took place soon after the harvest. Over twice as many first births took place during the last quarter of the year — roughly nine to thirteen months after the end of the harvest, depending on the region and crop — as took place during the first quarter of the year. This pattern cannot be attributed merely to the fact that slaves had more leisure time during the winter interstice, and hence, more opportunity for sexual intercourse. If that was all that had been involved, the peaking of births during the last quarter of the year would have occurred not only for first children but for subsequent children as well.

Also fallacious is the contention that slave marriages, since they were arbitrarily dictated by masters, frequently produced odd age combinations — young men married to old women and vice versa. Figure 39 shows that most marriages were contracted among partners quite close in age. The average age difference between husband and wife was just three years. In almost all cases, the man was the same age or older than the woman. Reversals in this pattern were quite uncommon.

That slave life pivoted around stable, nuclear families does not mean that the black family was merely a copy of the white family. No doubt the African heritage of blacks, as well as their particular socioeconomic circumstances, re-

Figure 38

A Comparison of the Seasonal Distribution of First Births with that of All Other Births

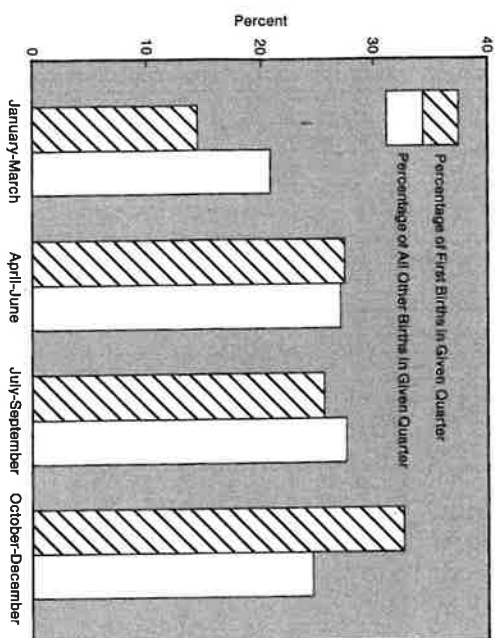
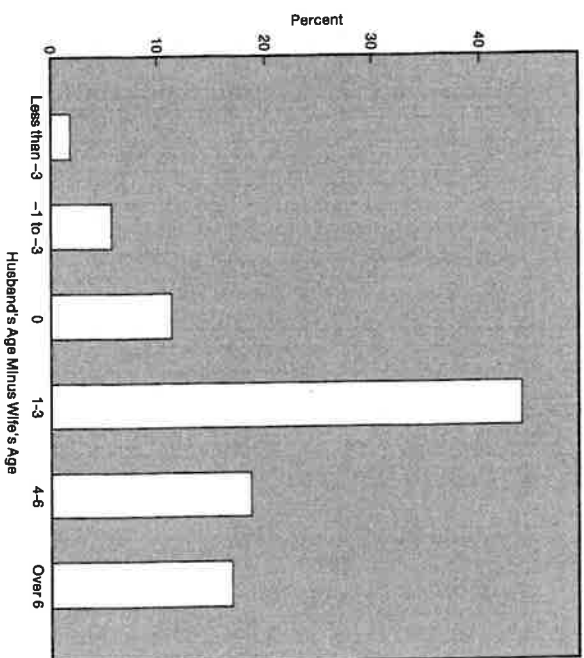


Figure 39

The Distribution of Age Differences between Slave Husbands and Wives



sulted in various characteristics which were, if not restricted to, at least more frequent among black than white families. For example, various bits of evidence suggest that wives tended to play a stronger role in black than in white families. Careful delineation of such special characteristics and the determination of their incidence is a task which has not yet been adequately essayed. The evidence already in hand, however, clearly invalidates many of the generalizations that now permeate history books.

It is not true that "the typical slave family was matriarchal in form" and that the "husband was at most his wife's assistant." Nor is it true that the "male slave's only crucial function within the family was that of siring offspring." For better or worse, the dominant role in slave society was played by men, not women. It was men who occupied virtually all of the managerial slots available to slaves. There were very few female overseers or drivers. Men occupied nearly all the artisan crafts; among them were carpentry, coopering and blacksmithing. In the city of Charleston in 1848, for example, all of the 706 slave artisans were male.

While females worked along with males in the field, their role was strictly delimited. Much has been made of the women who worked in plow gangs. But such participation was quite uncommon. Plow gangs were confined almost exclusively to men, and predominantly to young men. During the period of cultivation, women worked along with older men and children in the hoe gangs, where strength was not so important a factor. Only at harvesttime were women the equal of "any man" since dexterity, not strength, was the crucial characteristic of successful cotton picking. Just as some jobs on the plantation were confined strictly to men, others were confined strictly to women. Men were virtually never spinners, weavers, seamstresses, or nurses. The differentiation between male and female roles continued into the domestic staffs of plantations, although the division was

somewhat less sharp. Gardeners and coachmen were jobs for males; laundresses and cooks, female jobs.

There was also a division of labor within the slave family, a division that began with courtship. It was the male who, at least on the surface, initiated the period of courtship. And it was the man who secured the permission of the planter to marry. After marriage, the tasks of cooking, ordinary household cleaning, laundering, and care of the children fell to the mother. Work in the garden patches of the slave household, hunting and fishing for extra food, and chopping wood were among the tasks of the father.

Planters recognized husbands as the head of the family. Slave families were listed in their record books with the husband at the top of the list. Houses were assigned by the names of husbands and the semiannual issues of clothing to families were made in the name of the husband. Garden patches were assigned to the husbands and the money earned from the sale of crops from these patches was held in his name. When slaves wanted advances of cash from these accounts, they were made to the men. Slave purchases of cloth and apparel (whether intended for men, women, or children) were charged against the names of the husbands, as were purchases of such other items as pails, pots, and special foods.

While both moral convictions and good business practice generally led planters to encourage the development of stable nuclear families, it would be a mistake to assume that the black family was purely, or even predominantly, the creation of white masters. The exact interplay of external and internal forces in shaping the black family is still unknown. But there is considerable evidence that the nuclear form was not merely imposed on slaves. Slaves apparently abandoned the African family forms because they did not satisfy the needs of blacks who lived and worked under conditions and in a society much different from those which their ancestors experienced. The nuclear family took root among

blacks because it did satisfy those needs. Witness to the meaning which the family held for slaves is given by the deep anguish which they usually expressed on those occasions when their families were rent apart on the auction block. "Well, dey took us on up dere to Memphis and we was sold jest like cattle," said Nancy Gardner, a former slave who lived in Oklahoma. "Dey sold me and ma together and dey sold pa and de boys together. Dey was sent to Mississippi and we was sent to Alabama. My pa, O how my ma was grieved to death about him! She didn't live long after dat."

During the relatively infrequent instances when economic forces led the planter to destroy, rather than to maintain slave families, the *independent* striving of slaves to maintain their families came into sharp focus. Mrs. Josie Jordan, an ex-slave from Tennessee, reported that her mother "had two children while belonging to Mister Clark and he wouldn't let them go with mammy and pappy. That's what caused her misery. Pappy tried to ease her mind but she jest kept a'crying for her babies, Ann and Reuban, till Mister Lowery got Clark to leave them visit with her once a month." Further testimony to this striving is given by the ads which planters placed in newspapers advertising for the capture of runaways. These ads frequently indicate the planter's belief that his slave was attempting to reunite with the family from which the slave had recently been removed.

The abolitionist position on the black family, which has been accepted so uncritically by historians, was strikingly inconsistent. To arouse sentiment against the slave system they accurately portrayed the deep anguish which was caused by the forced breakup of slave families while simultaneously arguing that slavery had robbed black families of all meaning. This latter view was given vivid expression in Fanny Kemble's journal. The relationship between slave parents and children, she wrote, was reduced to the "connection between the animal and its young." In her view

black families were stripped of "all the unspeakable tenderness and solemnity, all the rational, and all the spiritual grace and glory" which she associated with parenthood in upper-class English families. Under slavery, she concluded, parenthood became "mere breeding, bearing, sucking, and there an end." The anguish on the auction block as well as the struggle of blacks to reunite their severed families, both during and immediately after slavery, suggests that the love that permeated slave families eluded Fanny Kemble and most other white observers — perhaps because of a veil of racial and class biases which obscured their vision and prevented them from seeing the real content of black family life.

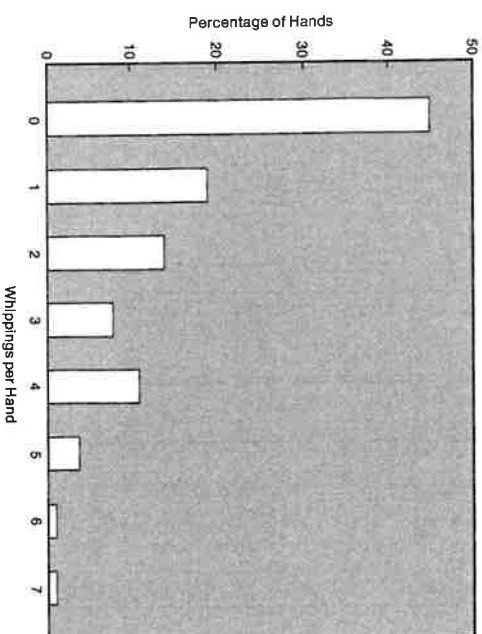
Punishment, Rewards, and Expropriation

The exploitative nature of slavery is most apparent in its system of punishment and rewards. Whipping was probably the most common punishment meted out against errant slaves. Other forms of punishment included the deprivation of various privileges (such as visits to town), confinement in stocks, incarceration, sale, branding, and the death penalty.

Whipping could be either a mild or a severe punishment, depending on how it was administered. Some whippings were so severe that they resulted in death. Indeed, in cases such as murder, the sentences of slaves who would otherwise have been executed were frequently converted to severe whipping, coupled with exportation to another state or a foreign country. For, as indicated in chapter 2, by converting the death penalty to whipping and exportation, the state could recover a substantial part of the value of a slave that

Figure 40

The Distribution of Whippings on the Bennet H. Barrow Plantation during a Two-Year Period Beginning in December, 1840



would have been lost through his execution. In other instances, whipping was as mildly applied as the corporal punishment normally practiced within families today.

Reliable data on the frequency of whipping is extremely sparse. The only systematic record of whipping now available for an extended period comes from the diary of Bennet Barrow, a Louisiana planter who believed that to spare the rod was to spoil the slave. His plantation numbered about 200 slaves, of whom about 120 were in the labor force. The record shows that over the course of two years a total of 160 whippings were administered, an average of 0.7 whippings per hand per year. About half the hands were not whipped at all during the period.

There was nothing exceptional about the use of whipping to enforce discipline among slaves until the beginning of the

nineteenth century. It must be remembered that through the centuries whipping was considered a fully acceptable form of punishment, not merely for criminals but also for honest men or women who in some way shirked their duties. Whipping of wives, for example, was even sanctified in some versions of the Scripture. The Matthew's Bible, which preceded the King James version, told the husband, in a note at 1 Pet. 3, that if his wife was "not obedient and heapfull vnto hym endeuoureth to beate the feare of God into her heade, and that thereby she maye be compelled to learne her duitie and do it." During the seventeenth and most of the eighteenth centuries whipping was commonly employed as a punishment in the North as well as in the South. Not until the end of the eighteenth century and the beginning of the nineteenth century did whipping rapidly fall from favor in the free states.

To attribute the continuation of whipping in the South to the maliciousness of masters is naïve. Although some masters were brutal, even sadistic, most were not. The overwhelming majority of the ex-slaves in the W.P.A. narratives who expressed themselves on the issue reported that their masters were good men. Such men worried about the proper role of whipping in a system of punishment and rewards. Some excluded it altogether. Most accepted it, but recognized that to be effective whipping had to be used with restraint and in a coolly calculated manner. Weston, for example, admonished his overseer not to impose punishment of any sort until twenty-four hours after the offense had been discovered. William J. Minor, a sugar planter, instructed his managers "not [to] cut the skin when punishing, nor punish in a passion." Many planters forbade the whipping of slaves except by them or in their presence. Others limited the number of lashes that could be administered without their permission.

The decline of whipping as an instrument of labor discipline outside of the South appears to have been heavily

influenced by economic considerations. With the rise of capitalism, impersonal and indirect sanctions were increasingly substituted for direct, personal ones. The hiring of free workers in the marketplace provided managers of labor with a powerful new disciplinary weapon. Workers who were lazy, indifferent, or who otherwise shirked their duties could be fired — left to starve beyond the eyesight or expense of the employer. Interestingly enough, denial of food was rarely used to enforce discipline on slaves. For the illness and lethargy caused by malnutrition reduced the capacity of the slave to labor in the fields. Planters preferred whipping to incarceration because the lash did not generally lead to an extended loss of the slave's labor time. In other words, whipping persisted in the South because the cost of substituting hunger and incarceration for the lash was greater for the slaveowner than for the northern employer of free labor. When the laborer owns his own human capital, forms of punishment which impair or diminish the value of that capital are borne exclusively by him. Under slavery, the master desired forms of punishment which, while they imposed costs on the slave, did so with minimum impairment to the human capital which the master owned. Whipping generally fulfilled these conditions.

While whipping was an integral part of the system of punishment and rewards, it was not the totality of the system. What planters wanted was not sullen and discontented slaves who did just enough to keep from getting whipped. They wanted devoted, hard-working, responsible slaves who identified their fortunes with the fortunes of their masters. Planters sought to imbue slaves with a "Protestant" work ethic and to transform that ethic from a state of mind into a high level of production. "My negroes have their name up in the neighbourhood," wrote Bennet Barrow, "for making more than any one else & they think whatever they do is better than any body Else." Such an attitude could not be beaten into slaves. It had to be elicited.

Much of the managerial attention of planters was focused on the problem of motivating their hands. To achieve the desired response they developed a wide-ranging system of rewards. Some rewards were directed toward improving short-run performance. Included in this category were prizes for the individual or the gang with the best picking record on a given day or during a given week. The prizes were such items as clothing, tobacco, and whiskey; sometimes the prize was cash. Good immediate performance was also rewarded with unscheduled holidays or with trips to town on weekends. When slaves worked at times normally set aside for rest, they received extra pay — usually in cash and at the rate prevailing in the region for hired labor. Slaves who were performing well were permitted to work on their own account after normal hours at such tasks as making shingles or weaving baskets, articles which they could sell either to their masters or to farmers in the neighborhood.

Some rewards were directed at influencing behavior over periods of intermediate duration. The rewards in this category were usually paid at the end of the year. Year-end bonuses, given either in goods or cash, were frequently quite substantial. Bennet Barrow, for example, distributed gifts averaging between \$15 and \$20 per slave family in both 1839 and 1840. The amounts received by particular slaves were proportional to their performance. It should be noted that \$20 was about a fifth of national per capita income in 1840. A bonus of the same relative magnitude today would be in the neighborhood of \$1,000.

Masters also rewarded slaves who performed well with patches of land ranging up to a few acres for each family. Slaves grew marketable crops on these lands, the proceeds of which accrued to them. On the Texas plantation of Julian S. Devereux, slaves operating such land produced as much as two bales of cotton per patch. Devereux marketed their crop along with his own. In a good year some of the slaves earned in excess of \$100 per annum for their

families. Devereux set up accounts to which he credited the proceeds of the sales. Slaves drew on these accounts when they wanted cash or when they wanted Devereux to purchase clothing, pots, pans, tobacco, or similar goods for them.

Occasionally planters even devised elaborate schemes for profit sharing with their slaves. William Jemison, an Alabama planter, entered into the following agreement with his bondsmen.

[Y]ou shall have two thirds of the corn and cotton made on the plantation and as much of the wheat as will reward you for the sowing it. I also furnish you with provisions for this year. When your crop is gathered, one third is to be set aside for me. You are then to pay your overseer his part and pay me what I furnish, clothe yourselves, pay your own taxes and doctor's fee with all expenses of the farm. You are to be no expense to me, but render to me one third of the produce and what I have loaned you. You have the use of the stock and plantation tools. You are to return them as good as they are and the plantation to be kept in good repair, and what clear money you make shall be divided equally amongst you in a fair proportion agreeable to the services rendered by each hand. There will be an account of all lost time kept, and those that earn most shall have most.

There was a third category of rewards. These were of a long-term nature, often requiring the lapse of a decade or more before they paid off. Thus, slaves had the opportunity to rise within the social and economic hierarchy that existed under bondage. Field hands could become artisans or drivers. Artisans could be allowed to move from the plantation to town where they would hire themselves out. Drivers could move up to the position of head driver or overseer. Climbing the economic ladder brought not only social status, and sometimes more freedom; it also had significant payoffs in better housing, better clothing, and cash bonuses.

Little attention has hitherto been paid to the manner in which planters selected the slaves who were to become the

artisans and managers. In some cases boys were apprenticed to carpenters, blacksmiths, or some similar craftsmen when they were in their early teens, as was typically done with whites. For slaves, this appears to have been the exception rather than the rule. Analysis of occupational data derived from probate and plantation records reveals an unusual distribution of ages among slave artisans. Slaves in their twenties were substantially underrepresented, while slaves in their forties and fifties were overrepresented. This age pattern suggests that the selection of slaves for training in the crafts was frequently delayed until slaves reached their late twenties, or perhaps even into the thirties.

Normally this would be an uneconomical policy, since the earlier an investment is made in occupational training, the more years there are to reap the returns on that investment. Slavery altered this pattern by shifting the authority to determine occupational investments from the parents to the masters. In free societies, kinship is usually the primary basis for determining which members of the new generation are trained in skilled occupations. But the slaveholder lacked the vested interests of a parent. He could, therefore, treat entry into the skilled occupations as a prize that was to be claimed by the most deserving, regardless of family background. The extra effort put forth by young field hands who competed for these jobs appears to have more than offset the loss in returns due to the curtailed period over which the occupational investment was amortized. We do not mean to suggest that kinship played no role in the intergenerational transfer of skills among slaves. We merely wish to stress that its role was significantly reduced as compared with free society.

Another long-run reward was freedom through manumission. The chance of achieving this reward was, of course, quite low. Census data indicate that in 1850 the manumission rate was just 0.45 per thousand slaves. Manumission could be achieved either through the philanthropy of a

master or through an agreement which permitted a slave to buy himself out. Sometimes gifts of freedom were bestowed while the master was still alive. More often it was a bequest set forth in a will. Self-purchase involved arrangements under which slaves were permitted to purchase themselves with money that they earned from work on their own account, or in the case of skilled urban slaves, by increasing the share of income which the artisan paid to his master. Some skilled slaves were able to accumulate enough capital to purchase their freedom within a decade. For others the period extended to two decades or more. Little information is currently available on the prices at which such transactions were concluded. It is not known whether slaves involved in self-purchase were generally forced to pay a price in excess of their market value.

From the foregoing it is clear that slaves did not all live at a uniform level of income. The elaborate system of rewards erected by planters introduced substantial variation in the slave standard of living. Much work remains to be done before it will be possible to reconstruct with reasonable accuracy the full range of the slave income distribution. It has been possible, however, to estimate the "basic income" of slaves in 1850. This, together with some fragmentary evidence on the higher incomes of slaves, will at least suggest the range of income variation that prevailed.

By "basic income" we mean the value of the food, clothing, shelter, and medical care furnished to slaves. The average value of the expenditure on these items for an adult male in 1850 was about \$48.00. The most complete information on the extra earnings of field hands comes from several Texas plantations. The leading hands on these estates frequently earned between \$40 and \$110 per year above basic income through the sale of cotton and other products raised on their patches. This experience was not unique to Texas. On one Alabama plantation, eight hands produced cotton that earned them an average of \$71 each, with the high man

collecting \$96. On still another plantation the average extra earnings of the thirteen top hands was \$77. These scattered cases suggest that the ratio of high earnings to basic earnings among field hands was in the neighborhood of 2.5.

When the incomes of artisans are taken into account, the spread in slave earnings became still wider. The top incomes earned by craftsmen must have been several times basic income. This is implied by the high prices which artisans had to pay to buy themselves out. The average price of a prime-aged blacksmith was about \$1,700 in 1850. Thus, a thirty-year-old man who was able to buy himself out in a decade probably earned in the neighborhood of \$170 per year over subsistence. This suggests a ratio of artisan to basic income of about 4.5.

The highest annual figure we have been able to uncover for extra earnings by a field hand in a single year is \$309. Aham, the Alabama slave whose sales of peaches, apples, and cotton yielded this sum, had accumulated enough capital over the years so that in 1860 he held notes on loans totaling over \$2,400. The ratio of Aham's agricultural income to basic income is 7.4. If we assume that Aham earned 6 percent, or \$144, on the loans, the ratio would rise to 10.4. The highest income above maintenance that we have found for an artisan is \$500. In this case the ratio of earned to basic income is 11.4.

While the reward structure created much more room for upward mobility within the slave system than is usually supposed, the scope of opportunity should not be exaggerated. The highest levels of attainment were irrevocably foreclosed to slaves. The entrepreneurial talent obviously possessed by bondsmen such as Aham could not be used to catapult them into the stewardship of great businesses as long as they remained slaves. No slave, regardless of his gifts, could aspire to political position. No man of letters — there were slaves who acquired considerable erudition — could ever hold an appointment in the faculty of a southern

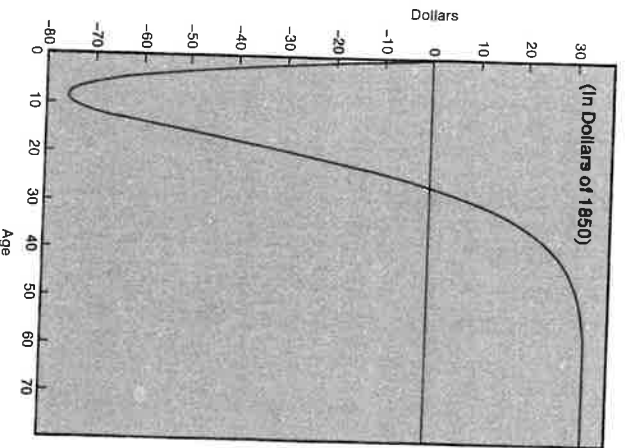
university as long as he was a bondsman. The entrepreneurial genius had to settle for lingering in the shadow of the master on whose protection he was dependent. The man of letters could go no further than the position of tutor to the children of a benevolent and enlightened planter. It was on the talented, the upper crust of slave society, that deprivations of the peculiar institution hung most heavy. This, perhaps, explains why it was that the first to flee to northern lines as Yankee advances corroded the Rebel positions were not the ordinary field hands, but the drivers and the artisans.

As previously noted, a part of the income that slaves produced was expropriated from them. Determination of the average annual amount of this expropriation over the life cycle of the entire slave population is an extremely complex matter. For the rate of expropriation differed from slave to slave, as well as from year to year for particular slaves. The issues involved in the computation and the procedures employed are discussed in appendix B. The results of the computation are displayed in figure 41. This figure presents the average accumulated value or, in the language of economists, the "expected present value" of the income that was expropriated at each age of the life cycle. Prior to age twenty-six, the accumulated expenditures by planters on slaves were greater than the average accumulated income which they took from them. After that age the reverse was true. Planters broke even early in the twenty-seventh year. Over the balance of the life cycle the accumulated or present value of the expropriation mounted, on average, to a total of \$32. This last figure is 12 percent of the average present value of the income earned by slaves over their lifetimes. In other words, on average, 12 percent of the value of the income produced by slaves was expropriated by their masters.

The relatively late age at which planters broke even is of great significance and requires further discussion. Two factors are responsible for this lateness. The first is that the

Figure 41

The Average Accumulated Value (Expected Present Value) of the Income Expropriated from Slaves over the Course of the Life Cycle



cost of capital was high in the South, and planters had to advance capital to cover the expense of rearing slaves for many years before they received a return from the labor of slaves. Second, because of the high mortality rates which prevailed for both black and white in the antebellum era, less than half the slaves lived to the break-even age. Fully 40 percent of the slaves died before age nineteen. Thus, a substantial part of the income taken from those slaves who survived into the later years was not an act of expropriation, but a payment required to cover the expenses of rearing children who failed to reach later ages. An additional part of the income taken from productive slaves, much smaller than that taken to cover child rearing, was used to sustain un-

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productive elderly slaves, as well as the incapacitated at all ages.

These intergenerational transfers of income were not, of course, limited to slave society. They took place among free men as well. For free men must also bear the cost of rearing the young, including those who fail to survive, as well as of supporting the sick and the aged. But among free men, the decision regarding the pattern of intergenerational transfers is made by parents rather than masters. The question thus arises, "How did masters alter the pattern of intergenerational transfers as compared to what it would have been if slaves had been free?" It is much easier to pose the question than to provide an answer. We do not yet have an answer to this question. We would, however, caution against two assumptions. It is not necessarily true that the substitution of planters for parents as decision makers on this issue had a large effect on the pattern of intergenerational transfers. Nor is it necessarily true that readers would consider all changes in the pattern of transfers made by the planters to have been negative. For example, planters spent more (a larger share of earnings) on the medical care of slave children than did the parents of freedmen during the decades following the Civil War.

The high break-even age also helps to explain why U.S. planters encouraged the fertility of slave women, while slaveowners in other parts of the hemisphere appear to have discouraged it. The crux of the matter is that child rearing was profitable only if the expected life of slaves at birth was greater than the break-even age. In the U.S., the life expectation of slaves exceeded the break-even age by more than a half decade. But in colonies such as Jamaica, available evidence suggests that life expectation fell below the break-even age, probably by at least a half decade.

Consequently, during most of the eighteenth century, masters in colonies such as Jamaica discouraged family

formation and high fertility rates, preferring to buy adult slaves in Africa rather than to rear them. It was general policy to maintain an imbalance between the sexes with men outnumbering women by a ratio of 4 to 3. To further reduce the basis for stable families, planters encouraged polygamy among slaves by rewarding favored men with second and third wives. Thus, for the remainder of the population, the male-female ratio was about 13 to 8. So large a disproportion between the sexes was bound to encourage sexual activity outside of the family and to reduce fertility. The care given to infants was quite poor compared with the South. Mothers were not encouraged to nurse their children; the southern practice of a 40 to 50 percent reduction of the workload of nursing mothers was not imitated in Jamaica. Further evidence of the neglect of pregnant women is to be found in the high rate of stillbirths. On one of the leading Jamaican plantations, 75 out of 345 births over a four-year period were stillborn. It has also been charged that abortion was both encouraged and widely practiced.

The 12 percent rate of expropriation reported on slave income falls well within the modern tax rate on workers. It has been estimated that about 30 percent of the income of workers at the poverty level is taken from them through sales, real estate, and income taxes. On the other hand, such workers, on average, receive payments and various services from the government which more than offset the tax burden. Were there any services received by slaves which offset the income expropriated from them? The answer is yes, but they cannot be quantified reliably at present. Slaves shared in the benefits of large-scale purchases made by the planter. Their clothing, for example, would have been more costly if purchased individually. Perhaps an even more important benefit was the saving on interest charges. Through the intervention of their masters slaves, in effect, were able to borrow at prime rates. Given the high interest charges which black sharecroppers suffered in the post-Civil War

era, and the debt-peonage which enmeshed so many of them, the level of interest rates is not a small issue. Pending a more precise measurement of offsetting services, it seems warranted to place the average net rate of the expropriation of slave income at about 10 percent.

Expropriation is not, however, the only form of exploitation. As is shown in chapter 6, the *economic* burden imposed on slaves by other forms of exploitation probably exceeded that due to expropriation by a considerable margin.